Unit 3 - DevOps and CICD

What is DevOps?

DevOps is a set of practices that combines software development (Dev) and IT operations (Ops) to shorten the development lifecycle and deliver features, fixes, and updates frequently in close alignment with business objectives. It emphasizes collaboration, automation, and continuous improvement.

DevOps Lifecycle

- Plan - Define features, requirements, and sprints
- Code - Write and review code using Git
- Build - Compile and package the application
- Test - Automated unit, integration, and security tests
- Release - Approve and prepare for deployment
- Deploy - Release to production environment
- Operate - Monitor and maintain the application
- Monitor - Collect metrics and feedback for next cycle

CI/CD Pipeline

Continuous Integration (CI)

Developers frequently merge code changes into a shared repository. Each merge triggers automated builds and tests to detect issues early.

Continuous Delivery (CD)

Extends CI by automatically deploying all code changes to staging environment after tests pass. Deployment to production is manual.

Continuous Deployment

Fully automated pipeline where every change that passes tests is automatically deployed to production with no human intervention.

AWS DevOps Tools

AWS CodeCommit

Managed Git repository service. Similar to GitHub but hosted on AWS.

AWS CodeBuild

Fully managed build service that compiles code, runs tests, and produces deployable packages.

AWS CodeDeploy

Automates application deployments to EC2, Lambda, or on-premises servers.

AWS CodePipeline

Orchestrates the entire CI/CD pipeline connecting all the tools together.

GitHub Actions

GitHub Actions is a CI/CD platform integrated into GitHub. Workflows are defined in YAML files in the .github/workflows directory. Actions are triggered by events like push, pull_request, or schedule. It is widely used for open source and commercial projects.

Docker and Containers

Docker packages applications and their dependencies into containers. Containers are lightweight, portable, and consistent across environments. Amazon ECS (Elastic Container Service) and EKS (Elastic Kubernetes Service) are AWS services for running containers at scale.